

## Per-scene F-7 NMI winner across Q-levels (LDA backbone)

Colour = recipe family from the wordification taxonomy. V20 (purple) emerges at Q=32 on Indian Pines, Salinas, Salinas-A, Pavia U, Botswana.

|      | Indian Pines              | Salinas                   | Salinas-A                 | Pavia U                   | KSC                       | Botswana                  |
|------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|---------------------------|
| Q=8  | <b>V20</b><br>NMI = 0.442 | <b>V12</b><br>NMI = 0.643 | <b>V3</b><br>NMI = 0.676  | <b>V12</b><br>NMI = 0.607 | <b>V3</b><br>NMI = 0.535  | <b>V8</b><br>NMI = 0.559  |
| Q=16 | <b>V20</b><br>NMI = 0.473 | <b>V12</b><br>NMI = 0.677 | <b>V2</b><br>NMI = 0.695  | <b>V3</b><br>NMI = 0.555  | <b>V3</b><br>NMI = 0.524  | <b>V8</b><br>NMI = 0.556  |
| Q=32 | <b>V2</b><br>NMI = 0.458  | <b>V12</b><br>NMI = 0.632 | <b>V20</b><br>NMI = 0.722 | <b>V20</b><br>NMI = 0.580 | <b>V20</b><br>NMI = 0.497 | <b>V20</b><br>NMI = 0.554 |

Pure spectral (V1-V3)

Differentiated / wavelet (V4-V6, V14)

Absorption / chemistry (V7, V8, V15)

Learnt codebook (V11-V13, V17)

Manifold (V18, V19)

Label-aware (V20)